DOCUMENTATION TECHNIQUE

Documentation Technique & Architecture du Pipeline

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CLI To-Do List Manager - Technical Specification & Architecture Document

1. Executive Summary & Architecture Overview

1.1 Executive Brief

The CLI To-Do List Manager is a Python-based command-line application designed for efficient task lifecycle management. The system utilizes a local JSON file for persistent storage and follows a layered architectural pattern consisting of a CLI dispatch layer, a service layer for business logic, and a storage layer for I/O operations. Core capabilities include task creation with sequential ID assignment, status updates, and flexible listing options.

1.2 Maturity Assessment

The project is structurally sound with a high health index, though it requires minor REFINEMENT. While the core implementation path is clearly defined, there is a lack of formal acceptance criteria for individual tasks and a need for defined security and performance constraints, specifically regarding JSON file size limits and input sanitization.

1.3 Technical Stack

- Python
- pyproject.toml

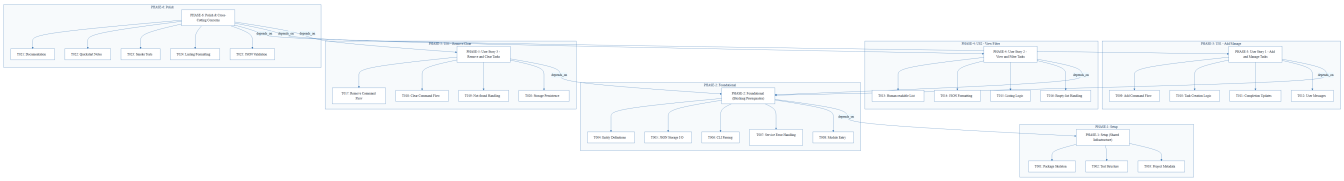
1.4 Architectural Constraints

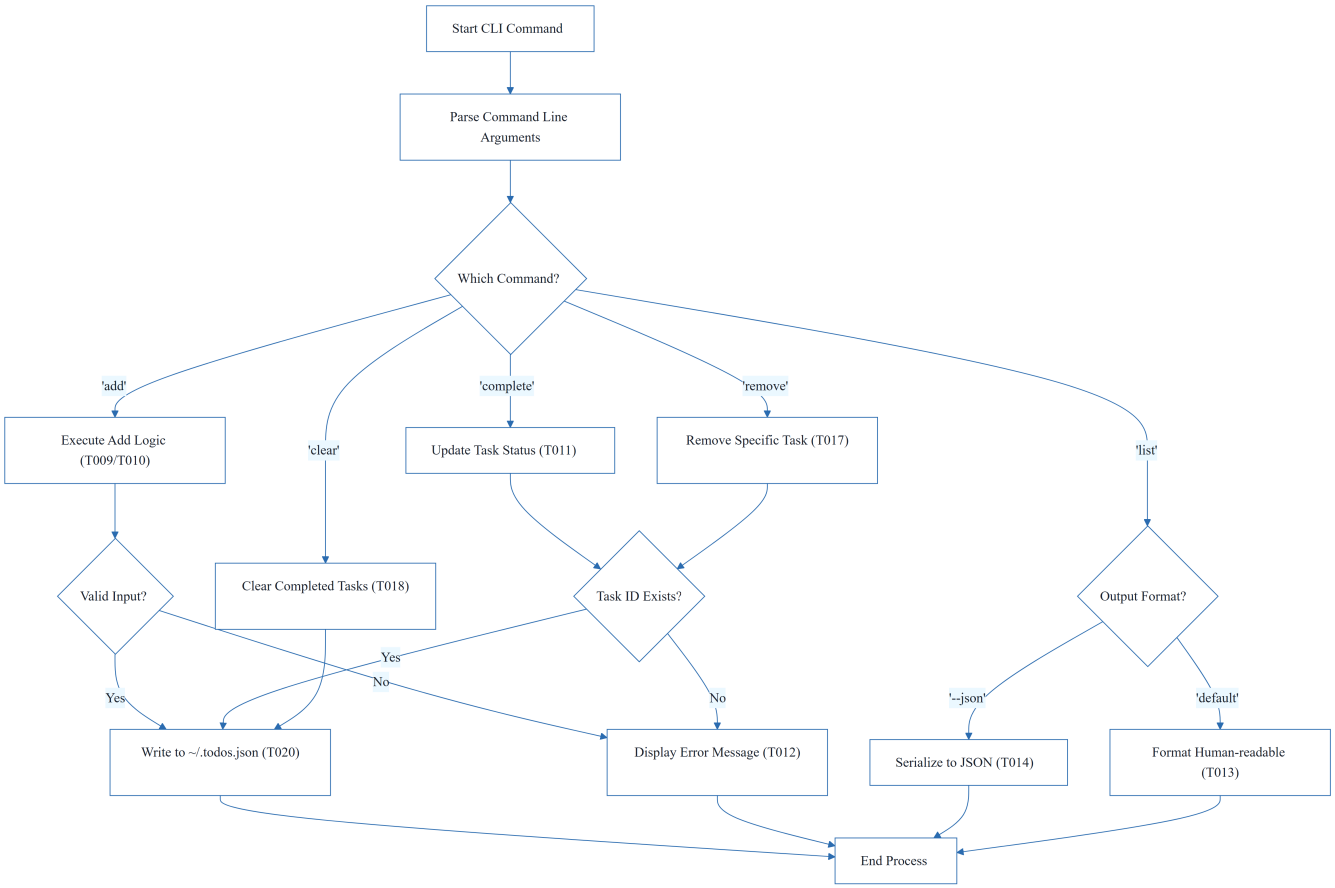
- Storage must be persisted in `~/.todos.json`.
- Task IDs must be assigned sequentially.
- Output must support both human-readable and raw JSON formats via the `--json` flag.
- Strict execution order: Setup $\rightarrow$ Foundational $\rightarrow$ User Stories $\rightarrow$ Polish.
- Service logic must be implemented prior to CLI output formatting.

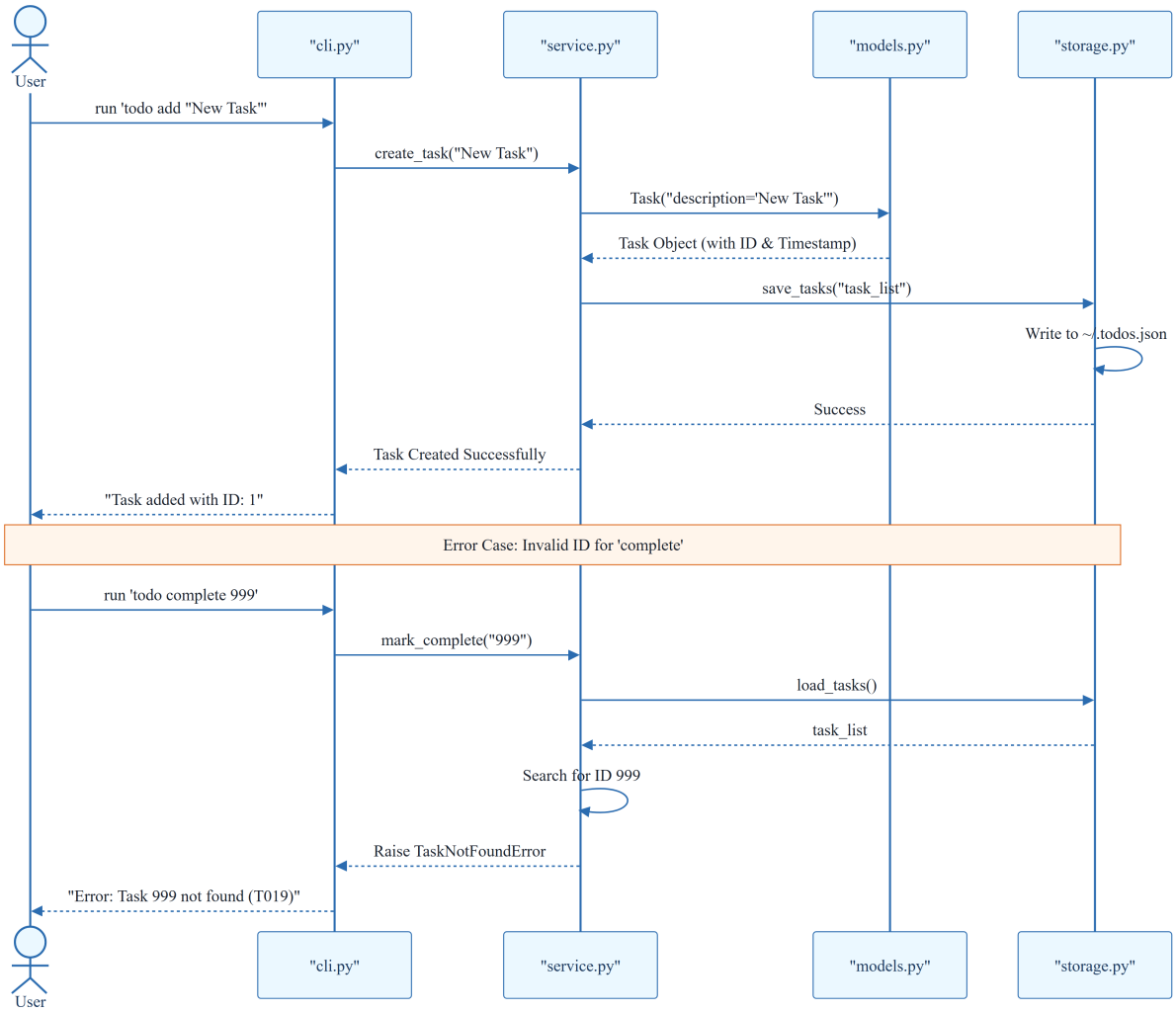
1.5 Critical Dependencies

- JSON file I/O for persistence in `~/.todos.json`.
- Sequential ID generation logic in `models.py`.
- Strict dependency of all User Stories on the Foundational Phase (PHASE-2).
- Referential integrity between CLI commands and service-layer task collection utilities.

2. Architecture Workflows

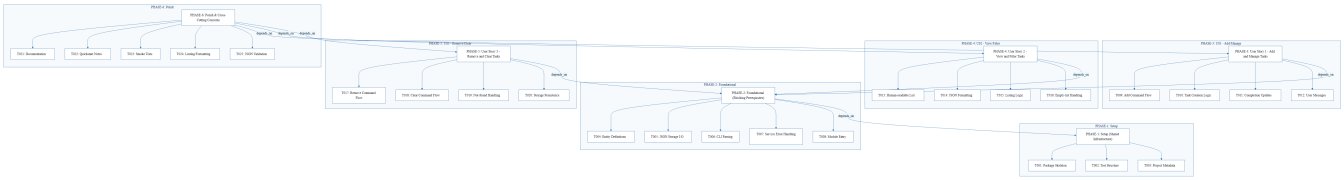




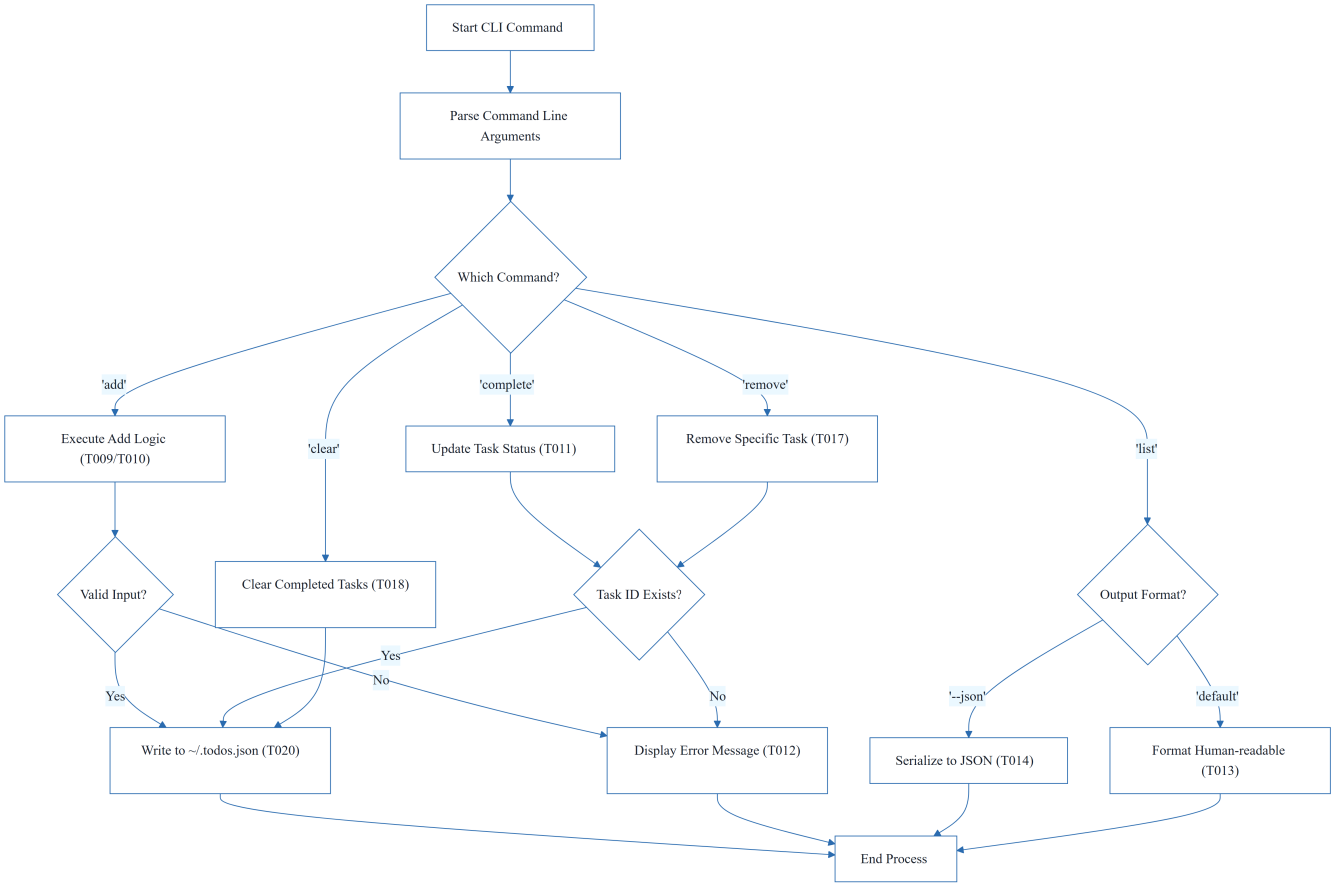


& Visual Diagrams

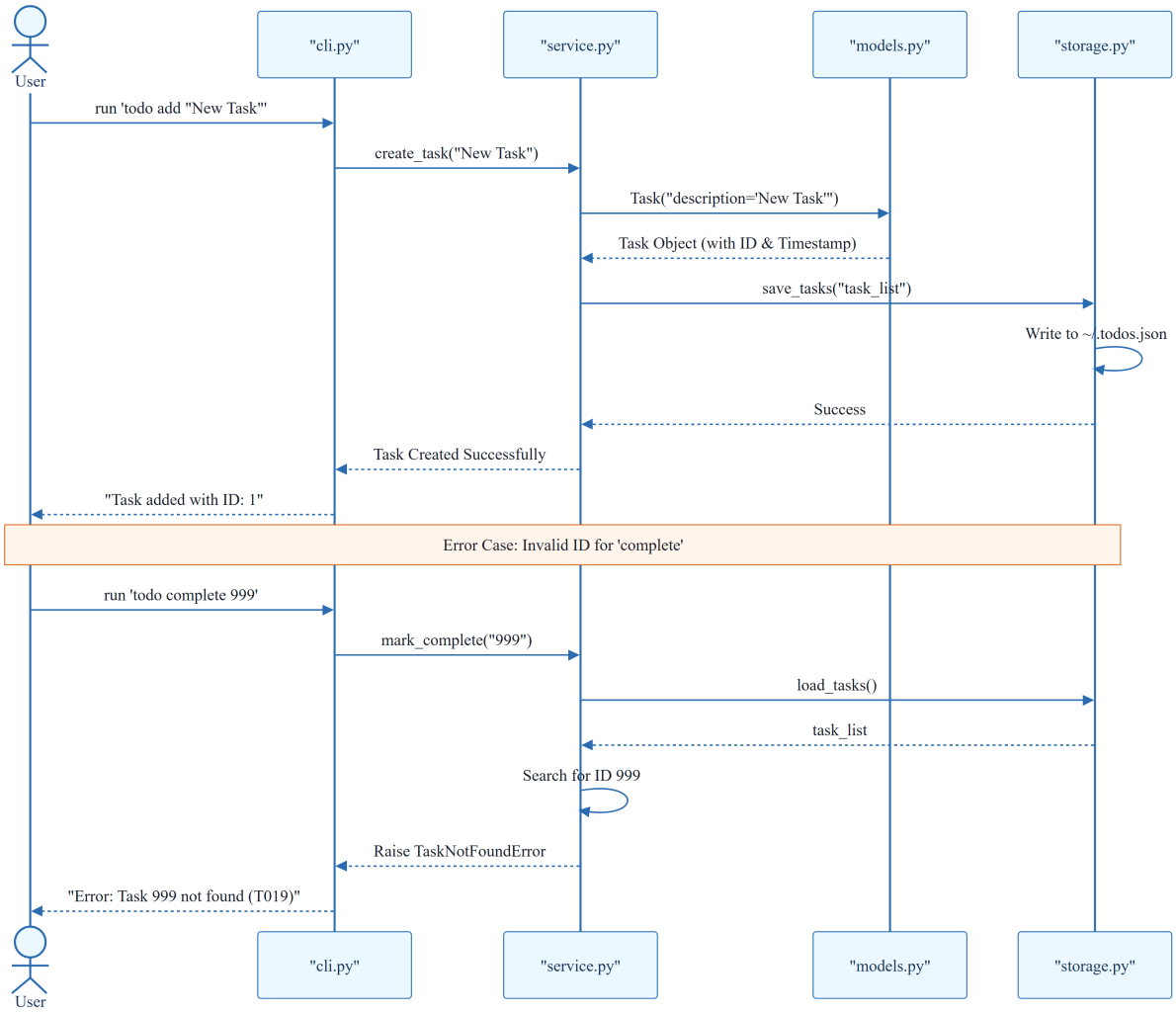
2.1 Project Implementation Roadmap & Traceability



2.2 CLI Command Execution Workflow



2.3 CLI To-Do Manager Sequence Interaction



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

ID	Requirement / Task Description	Phase	Story	Status
T001	Create Python package skeleton (`__init__.py`, `__main__.py`, `cli.py`, `models.py`, `storage.py`, `service.py`)	PHASE-1	N/A	Completed
T002	Create test directory structure (`tests/unit/`, `tests/integration/`)	PHASE-1	N/A	Completed

ID	Requirement / Task Description	Phase	Story	Status
T003	Add project metadata and tooling entry points in `pyproject.toml`	PHASE-1	N/A	Completed
T004	Implement task entity definitions and serialization helpers in `models.py`	PHASE-2	N/A	Completed
T005	Implement JSON storage path resolution and file I/O helpers in `storage.py`	PHASE-2	N/A	Completed
T006	Implement shared command-line parsing and top-level dispatch in `cli.py`	PHASE-2	N/A	Completed
T007	Implement shared service-layer error handling and task collection utilities in `service.py`	PHASE-2	N/A	Completed
T008	Define module entry behavior for `python -m todo_manager` in `__main__.py`	PHASE-2	N/A	Completed
T009	Implement the `add` command flow in `cli.py` and `service.py`	PHASE-3	US1	Completed
T010	Implement task creation, sequential ID assignment, and `created_at` population in `models.py`	PHASE-3	US1	Completed
T011	Implement completion updates for existing tasks in `service.py`	PHASE-3	US1	Completed
T012	Add user-facing success and error messages for add and complete operations in `cli.py`	PHASE-3	US1	Completed

ID	Requirement / Task Description	Phase	Story	Status
T013	Implement human-readable task listing output in `cli.py`	PHASE-4	US2	Completed
T014	Implement `--json` output formatting in `cli.py` using serialization helpers	PHASE-4	US2	Completed
T015	Add listing logic that preserves task order and status fields in `service.py`	PHASE-4	US2	Completed
T016	Handle the empty-list case with a clear message in `cli.py`	PHASE-4	US2	Completed
T017	Implement the `remove` command flow in `cli.py` and `service.py`	PHASE-5	US3	Pending
T018	Implement the `clear` command flow for removing completed tasks in `service.py`	PHASE-5	US3	Completed
T019	Add not-found handling for invalid task IDs in `cli.py`	PHASE-5	US3	Completed
T020	Ensure storage writes persist removals and clears safely in `storage.py`	PHASE-5	US3	Completed
T021	Document the CLI usage and storage behavior in `README.md`	PHASE-6	N/A	Completed
T022	Add quickstart verification notes and examples in `quickstart.md`	PHASE-6	N/A	Pending
T023	Run manual smoke test of `add`, `list`, `complete`, `remove`, and `clear`	PHASE-6	N/A	Pending

ID	Requirement / Task Description	Phase	Story	Status
T024	Verify linting and formatting expectations for source files	PHASE-6	N/A	Pending
T025	Confirm generated JSON output remains parseable for `todo list --json`	PHASE-6	N/A	Completed
TEST-US1	Validation: `todo add` and `todo complete` reflect in stored task list	PHASE-3	US1	N/A
TEST-US2	Validation: `todo list` and `todo list --json` output correctness	PHASE-4	US2	N/A
TEST-US3	Validation: `todo remove` and `todo clear` target correct entries	PHASE-5	US3	N/A

3.2 Security Rules

- **Input Sanitization**: (Gap identified) CLI inputs must be sanitized to prevent injection or malformed JSON writes.
- **File Permissions**: The application should ensure `~/.todos.json` is accessible only by the current user.

3.3 Data Models

- **Task Entity**:
- `id`: Integer (Sequential)
- `description`: String
- `completed`: Boolean
- `created_at`: Timestamp (ISO 8601)
- **Storage Format**: JSON array of Task objects.

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Missing Section	Priority	Remediation Advice
Acceptance Criteria	MEDIUM	While 'Independent Tests' are provided, formal acceptance criteria for each task would improve validation.
Security & Performance Constraints	LOW	Define constraints for JSON file size or input sanitization for the CLI.
Open Questions & Uncertainties	LOW	No open questions were listed in the source document.

4.2 Remediation & Workflow

1. **Validation Enhancement**: Integrate formal acceptance criteria into the `T-series` tasks.
2. **Constraint Definition**: Establish a maximum file size for `~/.todos.json` to prevent performance degradation during I/O.
3. **Sanitization Layer**: Implement a validation utility in `service.py` to scrub CLI inputs before they reach the `Model` layer.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition
Checkpoint	TECHNICAL_STACK	PHASE-2	A synchronization gate ensuring all blocking infrastructure is verified before parallel feature development commences.
Foundational	TECHNICAL_STACK	PHASE-2	The core shared infrastructure layer providing essential serialization and I/O capabilities required by all subsequent features.
Goal	BUSINESS_DOMAIN	PHASE-3	The primary functional objective a user must achieve within a specific feature set.
ID	TECHNICAL_STACK	Format: `[ID] [P?] [Story] Description`	A unique alphanumeric token used to track specific implementation units within the project roadmap.
JSON	TECHNICAL_STACK	T005	The lightweight data-interchange format used for persistent storage and machine-readable output.

Term	Category	Context Anchor	Project Definition
MVP	BUSINESS_DOMAIN	PHASE-3	The minimum set of functional capabilities required to provide basic value, specifically adding and completing entries.
Organization	BUSINESS_DOMAIN	Tasks: CLI To-Do List Manager	The structural grouping of implementation units by user story to ensure independent testability.
Prerequisites	TECHNICAL_STACK	Tasks: CLI To-Do List Manager	The set of design and research documents that must be available before implementation begins.
README	TECHNICAL_STACK	T021	The primary documentation file detailing usage instructions and storage behavior.
Setup	TECHNICAL_STACK	PHASE-1	The initial phase involving package skeleton creation and tooling configuration.
T016	TECHNICAL_STACK	T016	The specific implementation unit responsible for providing user feedback when no entries are available for display.
Tests	TECHNICAL_STACK	Tasks: CLI To-Do List Manager	The validation mechanisms, including unit, integration, and smoke verification, used to ensure system correctness.
population in	TECHNICAL_STACK	T010	The process of assigning a timestamp to the creation field during entry instantiation.
todo clear	BUSINESS_DOMAIN	TEST-US3	The operation that removes all entries marked as finished from the persistent store.
todo list	BUSINESS_DOMAIN	TEST-US2	The operation that retrieves and displays all stored entries in either human-readable or structured formats.
■■ CRITICAL	TECHNICAL_STACK	PHASE-2	A high-priority constraint indicating a hard dependency that blocks all subsequent feature work.